

DL_00_indice

March 13, 2026

1 Deep Learning con PyTorch

1.1 Índice del Curso

Bienvenido al curso de Deep Learning. Este material cubre desde los fundamentos teóricos hasta implementaciones prácticas con PyTorch.

1.2 Contenido Teórico

Notebook	Tema	Descripción
DL_01_fundamentos.ipynb	Fundamentos	Perceptrón, funciones de activación, backpropagation
DL_02_pytorch_basico.ipynb	PyTorch Básico	Tensores, autograd, módulos nn
DL_03_arquitecturas.ipynb	Arquitecturas	MLP, CNN, RNN/LSTM, Autoencoders
DL_04_entrenamiento.ipynb	Entrenamiento	Optimización, regularización, callbacks

1.3 Código Base

El directorio `02_fundamentos/core/` contiene las implementaciones:

- `modelos.py` - Arquitecturas: MLP, CNN, LSTM, Autoencoder, VAE
 - `entrenamiento.py` - Entrenador, EarlyStopping, schedulers
 - `utils.py` - Carga de datos, visualización, checkpointing
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1.4 Ejemplos Prácticos

El directorio `02_fundamentos/ejemplos/` contiene scripts ejecutables:

Script	Tarea	Modelo
<code>ejemplo_mlp_mnist.py</code>	Clasificación dígitos	MLP
<code>ejemplo_cnn_cifar10.py</code>	Clasificación imágenes	CNN
<code>ejemplo_lstm_sentimientos.py</code>	Análisis de texto	LSTM

1.5 Proyectos

El directorio `03_proyectos/` contiene proyectos completos:

1. **Clasificación de Imágenes** - Transfer learning con modelos preentrenados
 2. **NLP Sentimientos** - Análisis de sentimientos en reviews
 3. **Series Temporales** - Predicción con redes recurrentes
-

1.6 Requisitos

```
pip install -r requirements.txt
```

- Python 3.9+
- PyTorch 2.0+
- torchvision
- numpy, pandas, matplotlib
- scikit-learn
- tqdm

```
[1]: pip install -r requirements.txt
```

```
Collecting torch>=2.0.0 (from -r requirements.txt (line 2))
  Downloading torch-2.10.0-3-cp314-cp314-manylinux_2_28_x86_64.whl.metadata (31
kB)
Collecting torchvision>=0.15.0 (from -r requirements.txt (line 3))
  Downloading torchvision-0.25.0-cp314-cp314-manylinux_2_28_x86_64.whl.metadata
(5.4 kB)
Collecting torchaudio>=2.0.0 (from -r requirements.txt (line 4))
  Downloading torchaudio-2.10.0-cp314-cp314-manylinux_2_28_x86_64.whl.metadata
(6.7 kB)
Collecting numpy>=1.24.0 (from -r requirements.txt (line 5))
  Downloading
numpy-2.4.3-cp314-cp314-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata
(6.6 kB)
Collecting pandas>=2.0.0 (from -r requirements.txt (line 6))
  Downloading pandas-3.0.1-cp314-cp314-
manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (79 kB)
Collecting matplotlib>=3.7.0 (from -r requirements.txt (line 7))
  Downloading matplotlib-3.10.8-cp314-cp314-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (52 kB)
Collecting seaborn>=0.12.0 (from -r requirements.txt (line 8))
  Downloading seaborn-0.13.2-py3-none-any.whl.metadata (5.4 kB)
Collecting scikit-learn>=1.3.0 (from -r requirements.txt (line 9))
  Downloading scikit_learn-1.8.0-cp314-cp314-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (11 kB)
Collecting tqdm>=4.65.0 (from -r requirements.txt (line 10))
  Downloading tqdm-4.67.3-py3-none-any.whl.metadata (57 kB)
Requirement already satisfied: jupyter>=1.0.0 in ./venv/lib64/python3.14/site-
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packages (from -r requirements.txt (line 11)) (1.1.1)
Requirement already satisfied: ipywidgets>=8.0.0 in
./venv/lib64/python3.14/site-packages (from -r requirements.txt (line 12))
(8.1.8)
Collecting tensorboard>=2.13.0 (from -r requirements.txt (line 13))
  Downloading tensorboard-2.20.0-py3-none-any.whl.metadata (1.8 kB)
Collecting Pillow>=9.5.0 (from -r requirements.txt (line 14))
  Downloading pillow-12.1.1-cp314-cp314-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (8.8 kB)
Collecting filelock (from torch>=2.0.0->-r requirements.txt (line 2))
  Downloading filelock-3.25.2-py3-none-any.whl.metadata (2.0 kB)
Requirement already satisfied: typing-extensions>=4.10.0 in
./venv/lib64/python3.14/site-packages (from torch>=2.0.0->-r requirements.txt
(line 2)) (4.15.0)
Requirement already satisfied: setuptools in ./venv/lib64/python3.14/site-
packages (from torch>=2.0.0->-r requirements.txt (line 2)) (82.0.0)
Collecting sympy>=1.13.3 (from torch>=2.0.0->-r requirements.txt (line 2))
  Downloading sympy-1.14.0-py3-none-any.whl.metadata (12 kB)
Collecting networkx>=2.5.1 (from torch>=2.0.0->-r requirements.txt (line 2))
  Downloading networkx-3.6.1-py3-none-any.whl.metadata (6.8 kB)
Requirement already satisfied: Jinja2 in ./venv/lib64/python3.14/site-packages
(from torch>=2.0.0->-r requirements.txt (line 2)) (3.1.6)
Collecting fsspec>=0.8.5 (from torch>=2.0.0->-r requirements.txt (line 2))
  Downloading fsspec-2026.2.0-py3-none-any.whl.metadata (10 kB)
Collecting cuda-bindings==12.9.4 (from torch>=2.0.0->-r requirements.txt (line
2))
  Downloading cuda_bindings-12.9.4-cp314-cp314-
manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (2.6 kB)
Collecting nvidia-cuda-nvrtc-cu12==12.8.93 (from torch>=2.0.0->-r
requirements.txt (line 2))
  Downloading nvidia_cuda_nvrtc_cu12-12.8.93-py3-none-
manylinux2010_x86_64.manylinux_2_12_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cuda-runtime-cu12==12.8.90 (from torch>=2.0.0->-r
requirements.txt (line 2))
  Downloading nvidia_cuda_runtime_cu12-12.8.90-py3-none-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cuda-cupti-cu12==12.8.90 (from torch>=2.0.0->-r
requirements.txt (line 2))
  Downloading nvidia_cuda_cupti_cu12-12.8.90-py3-none-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cudnn-cu12==9.10.2.21 (from torch>=2.0.0->-r requirements.txt
(line 2))
  Downloading nvidia_cudnn_cu12-9.10.2.21-py3-none-
manylinux_2_27_x86_64.whl.metadata (1.8 kB)
Collecting nvidia-cublas-cu12==12.8.4.1 (from torch>=2.0.0->-r requirements.txt
(line 2))
  Downloading nvidia_cublas_cu12-12.8.4.1-py3-none-
manylinux_2_27_x86_64.whl.metadata (1.7 kB)

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Collecting nvidia-cufft-cu12==11.3.3.83 (from torch>=2.0.0->-r requirements.txt
(line 2))
  Downloading nvidia_cufft_cu12-11.3.3.83-py3-none-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-curand-cu12==10.3.9.90 (from torch>=2.0.0->-r requirements.txt
(line 2))
  Downloading nvidia_curand_cu12-10.3.9.90-py3-none-
manylinux_2_27_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cusolver-cu12==11.7.3.90 (from torch>=2.0.0->-r
requirements.txt (line 2))
  Downloading nvidia_cusolver_cu12-11.7.3.90-py3-none-
manylinux_2_27_x86_64.whl.metadata (1.8 kB)
Collecting nvidia-cuspars-cu12==12.5.8.93 (from torch>=2.0.0->-r
requirements.txt (line 2))
  Downloading nvidia_cuspars-cu12-12.5.8.93-py3-none-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (1.8 kB)
Collecting nvidia-cusparselt-cu12==0.7.1 (from torch>=2.0.0->-r requirements.txt
(line 2))
  Downloading nvidia_cusparselt_cu12-0.7.1-py3-none-
manylinux2014_x86_64.whl.metadata (7.0 kB)
Collecting nvidia-nccl-cu12==2.27.5 (from torch>=2.0.0->-r requirements.txt
(line 2))
  Downloading nvidia_nccl_cu12-2.27.5-py3-none-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (2.0 kB)
Collecting nvidia-nvshmem-cu12==3.4.5 (from torch>=2.0.0->-r requirements.txt
(line 2))
  Downloading nvidia_nvshmem_cu12-3.4.5-py3-none-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (2.1 kB)
Collecting nvidia-nvtx-cu12==12.8.90 (from torch>=2.0.0->-r requirements.txt
(line 2))
  Downloading nvidia_nvtx_cu12-12.8.90-py3-none-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (1.8 kB)
Collecting nvidia-nvjitlink-cu12==12.8.93 (from torch>=2.0.0->-r
requirements.txt (line 2))
  Downloading nvidia_nvjitlink_cu12-12.8.93-py3-none-
manylinux2010_x86_64.manylinux_2_12_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cufile-cu12==1.13.1.3 (from torch>=2.0.0->-r requirements.txt
(line 2))
  Downloading nvidia_cufile_cu12-1.13.1.3-py3-none-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (1.7 kB)
Collecting triton==3.6.0 (from torch>=2.0.0->-r requirements.txt (line 2))
  Downloading triton-3.6.0-cp314-cp314-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (1.7 kB)
Collecting cuda-pathfinder~=1.1 (from cuda-bindings==12.9.4->torch>=2.0.0->-r
requirements.txt (line 2))
  Downloading cuda_pathfinder-1.4.2-py3-none-any.whl.metadata (1.9 kB)
Requirement already satisfied: python-dateutil>=2.8.2 in
./venv/lib64/python3.14/site-packages (from pandas>=2.0.0->-r requirements.txt

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(line 6)) (2.9.0.post0)
Collecting contourpy>=1.0.1 (from matplotlib>=3.7.0->-r requirements.txt (line
7))
  Downloading contourpy-1.3.3-cp314-cp314-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (5.5 kB)
Collecting cyclor>=0.10 (from matplotlib>=3.7.0->-r requirements.txt (line 7))
  Downloading cyclor-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.22.0 (from matplotlib>=3.7.0->-r requirements.txt (line
7))
  Downloading fonttools-4.62.1-cp314-cp314-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (117 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib>=3.7.0->-r requirements.txt (line
7))
  Downloading kiwisolver-1.5.0-cp314-cp314-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (5.1 kB)
Requirement already satisfied: packaging>=20.0 in ./venv/lib64/python3.14/site-
packages (from matplotlib>=3.7.0->-r requirements.txt (line 7)) (26.0)
Collecting pyparsing>=3 (from matplotlib>=3.7.0->-r requirements.txt (line 7))
  Downloading pyparsing-3.3.2-py3-none-any.whl.metadata (5.8 kB)
Collecting scipy>=1.10.0 (from scikit-learn>=1.3.0->-r requirements.txt (line
9))
  Downloading scipy-1.17.1-cp314-cp314-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (62 kB)
Collecting joblib>=1.3.0 (from scikit-learn>=1.3.0->-r requirements.txt (line
9))
  Downloading joblib-1.5.3-py3-none-any.whl.metadata (5.5 kB)
Collecting threadpoolctl>=3.2.0 (from scikit-learn>=1.3.0->-r requirements.txt
(line 9))
  Downloading threadpoolctl-3.6.0-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: notebook in ./venv/lib64/python3.14/site-packages
(from jupyter>=1.0.0->-r requirements.txt (line 11)) (7.5.3)
Requirement already satisfied: jupyter-console in ./venv/lib64/python3.14/site-
packages (from jupyter>=1.0.0->-r requirements.txt (line 11)) (6.6.3)
Requirement already satisfied: nbconvert in ./venv/lib64/python3.14/site-
packages (from jupyter>=1.0.0->-r requirements.txt (line 11)) (7.17.0)
Requirement already satisfied: ipykernel in ./venv/lib64/python3.14/site-
packages (from jupyter>=1.0.0->-r requirements.txt (line 11)) (7.2.0)
Requirement already satisfied: jupyterlab in ./venv/lib64/python3.14/site-
packages (from jupyter>=1.0.0->-r requirements.txt (line 11)) (4.5.4)
Requirement already satisfied: comm>=0.1.3 in ./venv/lib64/python3.14/site-
packages (from ipywidgets>=8.0.0->-r requirements.txt (line 12)) (0.2.3)
Requirement already satisfied: ipython>=6.1.0 in ./venv/lib64/python3.14/site-
packages (from ipywidgets>=8.0.0->-r requirements.txt (line 12)) (9.10.0)
Requirement already satisfied: traitlets>=4.3.1 in ./venv/lib64/python3.14/site-
packages (from ipywidgets>=8.0.0->-r requirements.txt (line 12)) (5.14.3)
Requirement already satisfied: widgetsnbextension~=4.0.14 in
./venv/lib64/python3.14/site-packages (from ipywidgets>=8.0.0->-r
requirements.txt (line 12)) (4.0.15)

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Requirement already satisfied: jupyterlab_widgets~=3.0.15 in
./venv/lib64/python3.14/site-packages (from ipywidgets>=8.0.0->-r
requirements.txt (line 12)) (3.0.16)
Collecting absl-py>=0.4 (from tensorboard>=2.13.0->-r requirements.txt (line
13))
  Downloading absl_py-2.4.0-py3-none-any.whl.metadata (3.3 kB)
Collecting grpcio>=1.48.2 (from tensorboard>=2.13.0->-r requirements.txt (line
13))
  Downloading grpcio-1.78.0-cp314-cp314-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (3.8 kB)
Collecting markdown>=2.6.8 (from tensorboard>=2.13.0->-r requirements.txt (line
13))
  Downloading markdown-3.10.2-py3-none-any.whl.metadata (5.1 kB)
Collecting protobuf!=4.24.0,>=3.19.6 (from tensorboard>=2.13.0->-r
requirements.txt (line 13))
  Downloading protobuf-7.34.0-cp310-abi3-manylinux2014_x86_64.whl.metadata (595
bytes)
Collecting tensorboard-data-server<0.8.0,>=0.7.0 (from tensorboard>=2.13.0->-r
requirements.txt (line 13))
  Downloading tensorboard_data_server-0.7.2-py3-none-
manylinux_2_31_x86_64.whl.metadata (1.1 kB)
Collecting werkzeug>=1.0.1 (from tensorboard>=2.13.0->-r requirements.txt (line
13))
  Downloading werkzeug-3.1.6-py3-none-any.whl.metadata (4.0 kB)
Requirement already satisfied: decorator>=4.3.2 in ./venv/lib64/python3.14/site-
packages (from ipython>=6.1.0->ipywidgets>=8.0.0->-r requirements.txt (line 12))
(5.2.1)
Requirement already satisfied: ipython-pygments-lexers>=1.0.0 in
./venv/lib64/python3.14/site-packages (from
ipython>=6.1.0->ipywidgets>=8.0.0->-r requirements.txt (line 12)) (1.1.1)
Requirement already satisfied: jedi>=0.18.1 in ./venv/lib64/python3.14/site-
packages (from ipython>=6.1.0->ipywidgets>=8.0.0->-r requirements.txt (line 12))
(0.19.2)
Requirement already satisfied: matplotlib-inline>=0.1.5 in
./venv/lib64/python3.14/site-packages (from
ipython>=6.1.0->ipywidgets>=8.0.0->-r requirements.txt (line 12)) (0.2.1)
Requirement already satisfied: pexpect>4.3 in ./venv/lib64/python3.14/site-
packages (from ipython>=6.1.0->ipywidgets>=8.0.0->-r requirements.txt (line 12))
(4.9.0)
Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in
./venv/lib64/python3.14/site-packages (from
ipython>=6.1.0->ipywidgets>=8.0.0->-r requirements.txt (line 12)) (3.0.52)
Requirement already satisfied: pygments>=2.11.0 in ./venv/lib64/python3.14/site-
packages (from ipython>=6.1.0->ipywidgets>=8.0.0->-r requirements.txt (line 12))
(2.19.2)
Requirement already satisfied: stack_data>=0.6.0 in
./venv/lib64/python3.14/site-packages (from
ipython>=6.1.0->ipywidgets>=8.0.0->-r requirements.txt (line 12)) (0.6.3)

```

Requirement already satisfied: wcwidth in ./venv/lib64/python3.14/site-packages (from prompt_toolkit<3.1.0,>=3.0.41->ipython>=6.1.0->ipywidgets>=8.0.0->-r requirements.txt (line 12)) (0.6.0)

Requirement already satisfied: parso<0.9.0,>=0.8.4 in ./venv/lib64/python3.14/site-packages (from jedi>=0.18.1->ipython>=6.1.0->ipywidgets>=8.0.0->-r requirements.txt (line 12)) (0.8.6)

Requirement already satisfied: ptyprocess>=0.5 in ./venv/lib64/python3.14/site-packages (from pexpect>4.3->ipython>=6.1.0->ipywidgets>=8.0.0->-r requirements.txt (line 12)) (0.7.0)

Requirement already satisfied: six>=1.5 in ./venv/lib64/python3.14/site-packages (from python-dateutil>=2.8.2->pandas>=2.0.0->-r requirements.txt (line 6)) (1.17.0)

Requirement already satisfied: executing>=1.2.0 in ./venv/lib64/python3.14/site-packages (from stack_data>=0.6.0->ipython>=6.1.0->ipywidgets>=8.0.0->-r requirements.txt (line 12)) (2.2.1)

Requirement already satisfied: asttokens>=2.1.0 in ./venv/lib64/python3.14/site-packages (from stack_data>=0.6.0->ipython>=6.1.0->ipywidgets>=8.0.0->-r requirements.txt (line 12)) (3.0.1)

Requirement already satisfied: pure-eval in ./venv/lib64/python3.14/site-packages (from stack_data>=0.6.0->ipython>=6.1.0->ipywidgets>=8.0.0->-r requirements.txt (line 12)) (0.2.3)

Collecting mpmath<1.4,>=1.1.0 (from sympy>=1.13.3->torch>=2.0.0->-r requirements.txt (line 2))

Downloading mpmath-1.3.0-py3-none-any.whl.metadata (8.6 kB)

Requirement already satisfied: markupsafe>=2.1.1 in ./venv/lib64/python3.14/site-packages (from werkzeug>=1.0.1->tensorboard>=2.13.0->-r requirements.txt (line 13)) (3.0.3)

Requirement already satisfied: debugpy>=1.6.5 in ./venv/lib64/python3.14/site-packages (from ipykernel->jupyter>=1.0.0->-r requirements.txt (line 11)) (1.8.20)

Requirement already satisfied: jupyter-client>=8.8.0 in ./venv/lib64/python3.14/site-packages (from ipykernel->jupyter>=1.0.0->-r requirements.txt (line 11)) (8.8.0)

Requirement already satisfied: jupyter-core!=6.0.*,>=5.1 in ./venv/lib64/python3.14/site-packages (from ipykernel->jupyter>=1.0.0->-r requirements.txt (line 11)) (5.9.1)

Requirement already satisfied: nest-asyncio>=1.4 in ./venv/lib64/python3.14/site-packages (from ipykernel->jupyter>=1.0.0->-r requirements.txt (line 11)) (1.6.0)

Requirement already satisfied: psutil>=5.7 in ./venv/lib64/python3.14/site-packages (from ipykernel->jupyter>=1.0.0->-r requirements.txt (line 11)) (7.2.2)

Requirement already satisfied: pyzmq>=25 in ./venv/lib64/python3.14/site-packages (from ipykernel->jupyter>=1.0.0->-r requirements.txt (line 11)) (27.1.0)

Requirement already satisfied: tornado>=6.4.1 in ./venv/lib64/python3.14/site-packages (from ipykernel->jupyter>=1.0.0->-r requirements.txt (line 11)) (6.5.4)

Requirement already satisfied: platformdirs>=2.5 in

./venv/lib64/python3.14/site-packages (from jupyter-core!=6.0.*,>=5.1->ipykernel->jupyter>=1.0.0->-r requirements.txt (line 11)) (4.7.0)

Requirement already satisfied: async-lru>=1.0.0 in ./venv/lib64/python3.14/site-packages (from jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (2.1.0)

Requirement already satisfied: httpx<1,>=0.25.0 in ./venv/lib64/python3.14/site-packages (from jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (0.28.1)

Requirement already satisfied: jupyter-lsp>=2.0.0 in ./venv/lib64/python3.14/site-packages (from jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (2.3.0)

Requirement already satisfied: jupyter-server<3,>=2.4.0 in ./venv/lib64/python3.14/site-packages (from jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (2.17.0)

Requirement already satisfied: jupyterlab-server<3,>=2.28.0 in ./venv/lib64/python3.14/site-packages (from jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (2.28.0)

Requirement already satisfied: notebook-shim>=0.2 in ./venv/lib64/python3.14/site-packages (from jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (0.2.4)

Requirement already satisfied: anyio in ./venv/lib64/python3.14/site-packages (from httpx<1,>=0.25.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (4.12.1)

Requirement already satisfied: certifi in ./venv/lib64/python3.14/site-packages (from httpx<1,>=0.25.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (2026.1.4)

Requirement already satisfied: httpcore==1.* in ./venv/lib64/python3.14/site-packages (from httpx<1,>=0.25.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (1.0.9)

Requirement already satisfied: idna in ./venv/lib64/python3.14/site-packages (from httpx<1,>=0.25.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (3.11)

Requirement already satisfied: h11>=0.16 in ./venv/lib64/python3.14/site-packages (from httpcore==1.*->httpx<1,>=0.25.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (0.16.0)

Requirement already satisfied: argon2-cffi>=21.1 in ./venv/lib64/python3.14/site-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (25.1.0)

Requirement already satisfied: jupyter-events>=0.11.0 in ./venv/lib64/python3.14/site-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (0.12.0)

Requirement already satisfied: jupyter-server-terminals>=0.4.4 in ./venv/lib64/python3.14/site-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (0.5.4)

Requirement already satisfied: nbformat>=5.3.0 in ./venv/lib64/python3.14/site-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (5.10.4)

Requirement already satisfied: prometheus-client>=0.9 in ./venv/lib64/python3.14/site-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (0.24.1)

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Requirement already satisfied: terminado>=0.8.3 in ./venv/lib64/python3.14/site-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (0.18.1)

Requirement already satisfied: websocket-client>=1.7 in ./venv/lib64/python3.14/site-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (1.9.0)

Requirement already satisfied: babel>=2.10 in ./venv/lib64/python3.14/site-packages (from jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (2.18.0)

Requirement already satisfied: json5>=0.9.0 in ./venv/lib64/python3.14/site-packages (from jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (0.13.0)

Requirement already satisfied: jsonschema>=4.18.0 in ./venv/lib64/python3.14/site-packages (from jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (4.26.0)

Requirement already satisfied: requests>=2.31 in ./venv/lib64/python3.14/site-packages (from jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (2.32.5)

Requirement already satisfied: argon2-cffi-bindings in ./venv/lib64/python3.14/site-packages (from argon2-cffi>=21.1->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (25.1.0)

Requirement already satisfied: attrs>=22.2.0 in ./venv/lib64/python3.14/site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (25.4.0)

Requirement already satisfied: jsonschema-specifications>=2023.03.6 in ./venv/lib64/python3.14/site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (2025.9.1)

Requirement already satisfied: referencing>=0.28.4 in ./venv/lib64/python3.14/site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (0.37.0)

Requirement already satisfied: rpds-py>=0.25.0 in ./venv/lib64/python3.14/site-

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 (0.30.0)
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 (4.0.0)
 Requirement already satisfied: pyyaml>=5.3 in ./venv/lib64/python3.14/site-
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 (6.0.3)
 Requirement already satisfied: rfc3339-validator in
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 (0.1.4)
 Requirement already satisfied: rfc3986-validator>=0.1.1 in
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 (0.1.1)
 Requirement already satisfied: fqdn in ./venv/lib64/python3.14/site-packages
 (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-
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 (1.5.1)
 Requirement already satisfied: isoduration in ./venv/lib64/python3.14/site-
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 requirements.txt (line 11)) (20.11.0)
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 requirements.txt (line 11)) (3.0.0)
 Requirement already satisfied: rfc3987-syntax>=1.1.0 in
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 (1.1.0)
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 requirements.txt (line 11)) (1.3.0)
 Requirement already satisfied: webcolors>=24.6.0 in
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 (25.10.0)
 Requirement already satisfied: beautifulsoup4 in ./venv/lib64/python3.14/site-
 packages (from nbconvert->jupyter>=1.0.0->-r requirements.txt (line 11))
 (4.14.3)

Requirement already satisfied: bleach!=5.0.0 in ./venv/lib64/python3.14/site-packages (from bleach[css]!=5.0.0->nbconvert->jupyter>=1.0.0->-r requirements.txt (line 11)) (6.3.0)

Requirement already satisfied: defusedxml in ./venv/lib64/python3.14/site-packages (from nbconvert->jupyter>=1.0.0->-r requirements.txt (line 11)) (0.7.1)

Requirement already satisfied: jupyterlab-pygments in ./venv/lib64/python3.14/site-packages (from nbconvert->jupyter>=1.0.0->-r requirements.txt (line 11)) (0.3.0)

Requirement already satisfied: mistune<4,>=2.0.3 in ./venv/lib64/python3.14/site-packages (from nbconvert->jupyter>=1.0.0->-r requirements.txt (line 11)) (3.2.0)

Requirement already satisfied: nbclient>=0.5.0 in ./venv/lib64/python3.14/site-packages (from nbconvert->jupyter>=1.0.0->-r requirements.txt (line 11)) (0.10.4)

Requirement already satisfied: pandocfilters>=1.4.1 in ./venv/lib64/python3.14/site-packages (from nbconvert->jupyter>=1.0.0->-r requirements.txt (line 11)) (1.5.1)

Requirement already satisfied: webencodings in ./venv/lib64/python3.14/site-packages (from bleach!=5.0.0->bleach[css]!=5.0.0->nbconvert->jupyter>=1.0.0->-r requirements.txt (line 11)) (0.5.1)

Requirement already satisfied: tinycss2<1.5,>=1.1.0 in ./venv/lib64/python3.14/site-packages (from bleach[css]!=5.0.0->nbconvert->jupyter>=1.0.0->-r requirements.txt (line 11)) (1.4.0)

Requirement already satisfied: fastjsonschema>=2.15 in ./venv/lib64/python3.14/site-packages (from nbformat>=5.3.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (2.21.2)

Requirement already satisfied: charset_normalizer<4,>=2 in ./venv/lib64/python3.14/site-packages (from requests>=2.31->jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (3.4.4)

Requirement already satisfied: urllib3<3,>=1.21.1 in ./venv/lib64/python3.14/site-packages (from requests>=2.31->jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (2.6.3)

Requirement already satisfied: lark>=1.2.2 in ./venv/lib64/python3.14/site-packages (from rfc3987-syntax>=1.1.0->jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (1.3.1)

Requirement already satisfied: cffi>=2.0.0b1 in ./venv/lib64/python3.14/site-packages (from argon2-cffi-bindings->argon2-cffi>=21.1->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (2.0.0)

Requirement already satisfied: pycparser in ./venv/lib64/python3.14/site-packages (from cffi>=2.0.0b1->argon2-cffi-bindings->argon2-cffi>=21.1->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11))

(3.0)

Requirement already satisfied: soupsieve>=1.6.1 in ./venv/lib64/python3.14/site-packages (from beautifulsoup4->nbconvert->jupyter>=1.0.0->-r requirements.txt (line 11)) (2.8.3)

Requirement already satisfied: arrow>=0.15.0 in ./venv/lib64/python3.14/site-packages (from isoduration->jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (1.4.0)

Requirement already satisfied: tzdata in ./venv/lib64/python3.14/site-packages (from arrow>=0.15.0->isoduration->jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r requirements.txt (line 11)) (2025.3)

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[torchaudio]rchaudio[rchvision]ver-cu12]2]

Successfully installed Pillow-12.1.1 absl-py-2.4.0 contourpy-1.3.3 cuda-bindings-12.9.4 cuda-pathfinder-1.4.2 cycler-0.12.1 filelock-3.25.2

fonttools-4.62.1 fsspec-2026.2.0 grpcio-1.78.0 joblib-1.5.3 kiwisolver-1.5.0

markdown-3.10.2 matplotlib-3.10.8 mpmath-1.3.0 networkx-3.6.1 numpy-2.4.3

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sympy-1.14.0 tensorboard-2.20.0 tensorboard-data-server-0.7.2

threadpoolctl-3.6.0 torch-2.10.0 torchaudio-2.10.0 torchvision-0.25.0

tqdm-4.67.3 triton-3.6.0 werkzeug-3.1.6

[notice] A new release of pip is available: 25.1.1 -> 26.0.1
[notice] To update, run:
`pip install --upgrade pip`
Note: you may need to restart the kernel to use updated packages.

```
[2]: # Verificar instalación
import torch
print(f"PyTorch version: {torch.__version__}")
print(f"CUDA disponible: {torch.cuda.is_available()}")
if torch.cuda.is_available():
    print(f"GPU: {torch.cuda.get_device_name(0)}")
```

PyTorch version: 2.10.0+cu128
CUDA disponible: False